Geotagged Airline Tweets

A clean Elasticsearch pipeline for full-text, aggregation, and geospatial analysis

PYTHON INDEXER | ELASTICSEARCH 9 | 14 KIBANA QUERIES

14,640	14,485	847	5.85%
CSV rows	unique documents	geotagged docs	location share

Dataset and indexing

The Twitter US Airline Sentiment dataset contains 14,640 rows. There are 1,019 non-empty coordinate rows, including 164 placeholder points equal to `[0.0, 0.0]`. The indexer accepts 855 valid coordinate rows. Because 155 tweet IDs repeat, Elasticsearch stores 14,485 unique documents; 847 of them retain a valid location after last-write-wins updates.

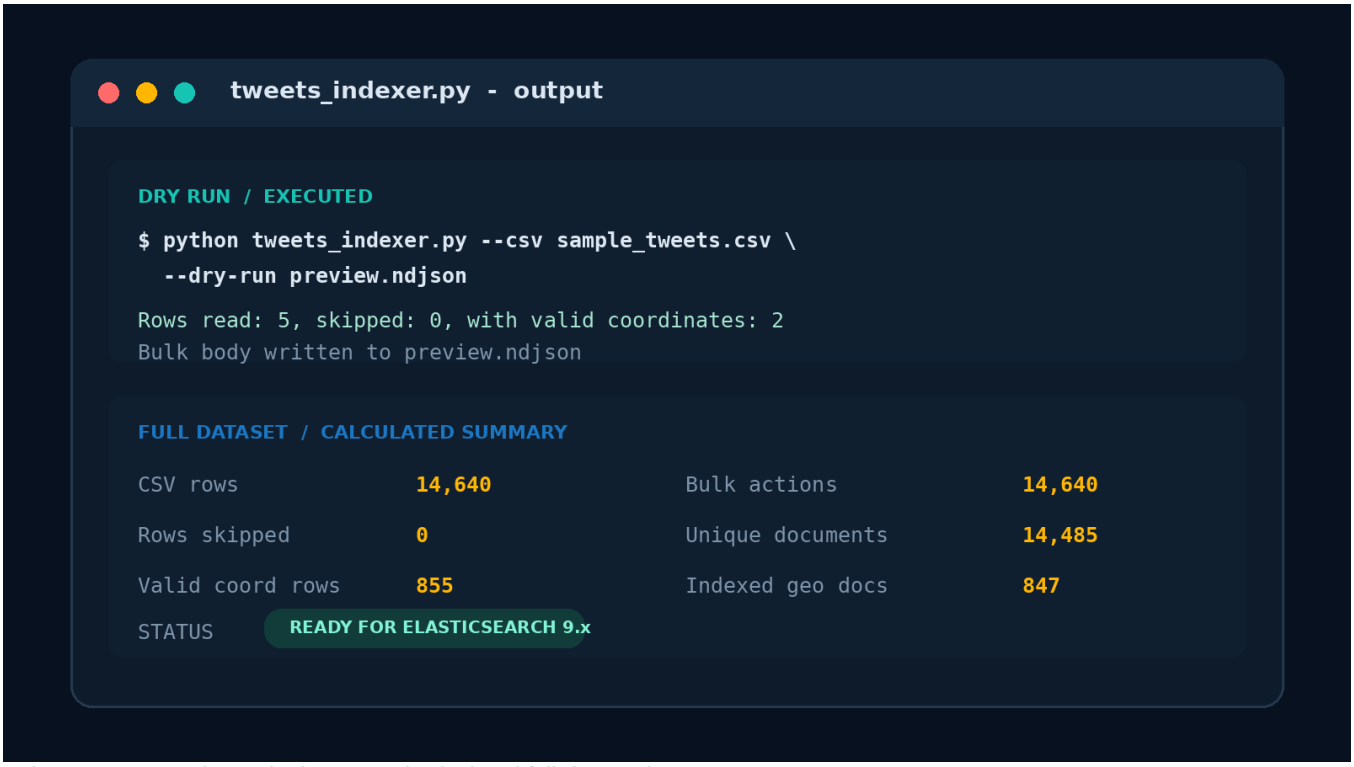


Figure 1. Executed sample dry-run and calculated full-dataset import summary.

Search and aggregation results

Q1-Q9 combine exact filters, analyzed text search, phrase matching, ranking, and bucket aggregations.

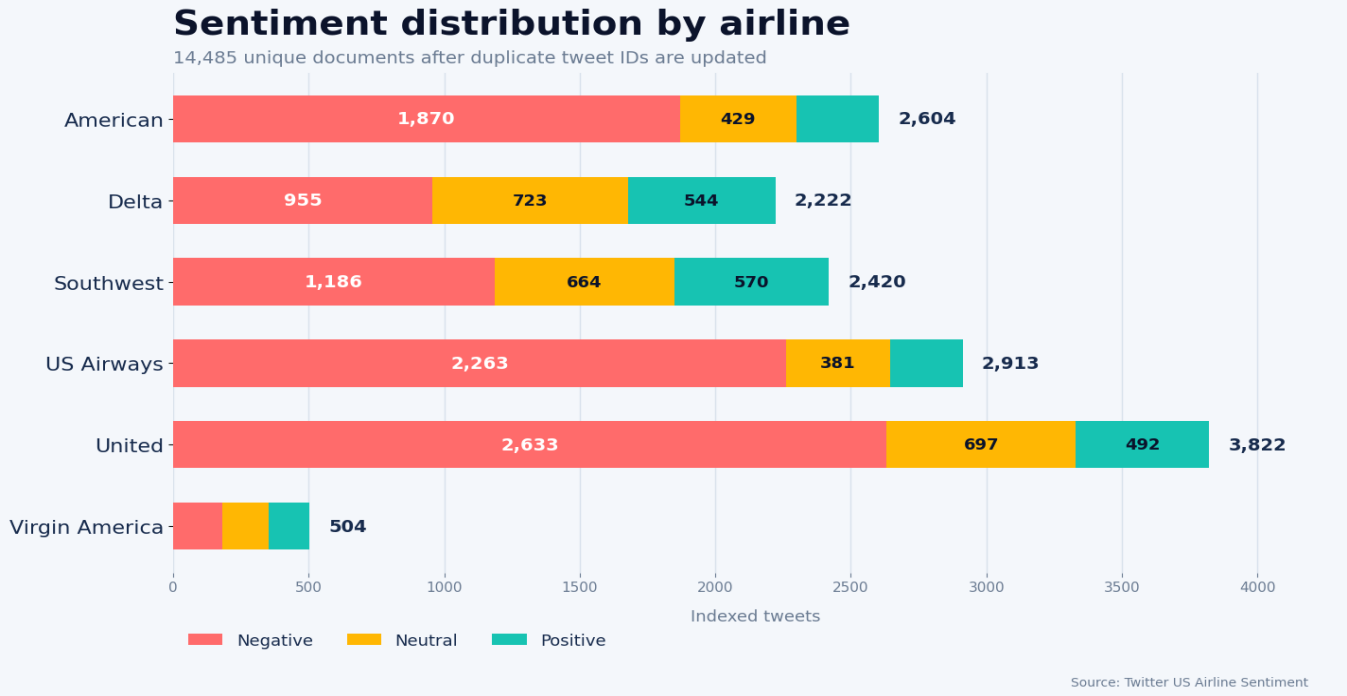


Figure 2. Sentiment breakdown by airline after duplicate tweet IDs are updated.

QUERY	KEY RESULT
Q1	9,088 negative tweets; 2,633 are about United.
Q2	1,925 matches for lost or delayed luggage and bags. Top examples include United, Southwest, and American.
Q3	90 positive tweets contain the exact phrase `customer service`.
Q4	25 negative Delta / Late Flight tweets on February 23, 2015, Pacific time.
Q5	The five highest retweet counts are 44, 32, 31, 28, and 22.

QUERY	KEY RESULT
Q6	United has the most documents (3,822); 2,633 are negative.
Q7	Customer Service Issue leads for five airlines; Late Flight leads for Delta.
Q8	February 22 is the busiest day with 3,073 tweets.
Q9	#destinationdragons leads with 81 documents, followed by #fail with 67.

Geospatial analysis

Q10-Q14 use geo filters, distance sorting, bounding boxes, distance rings, geotile cells, and a centroid.

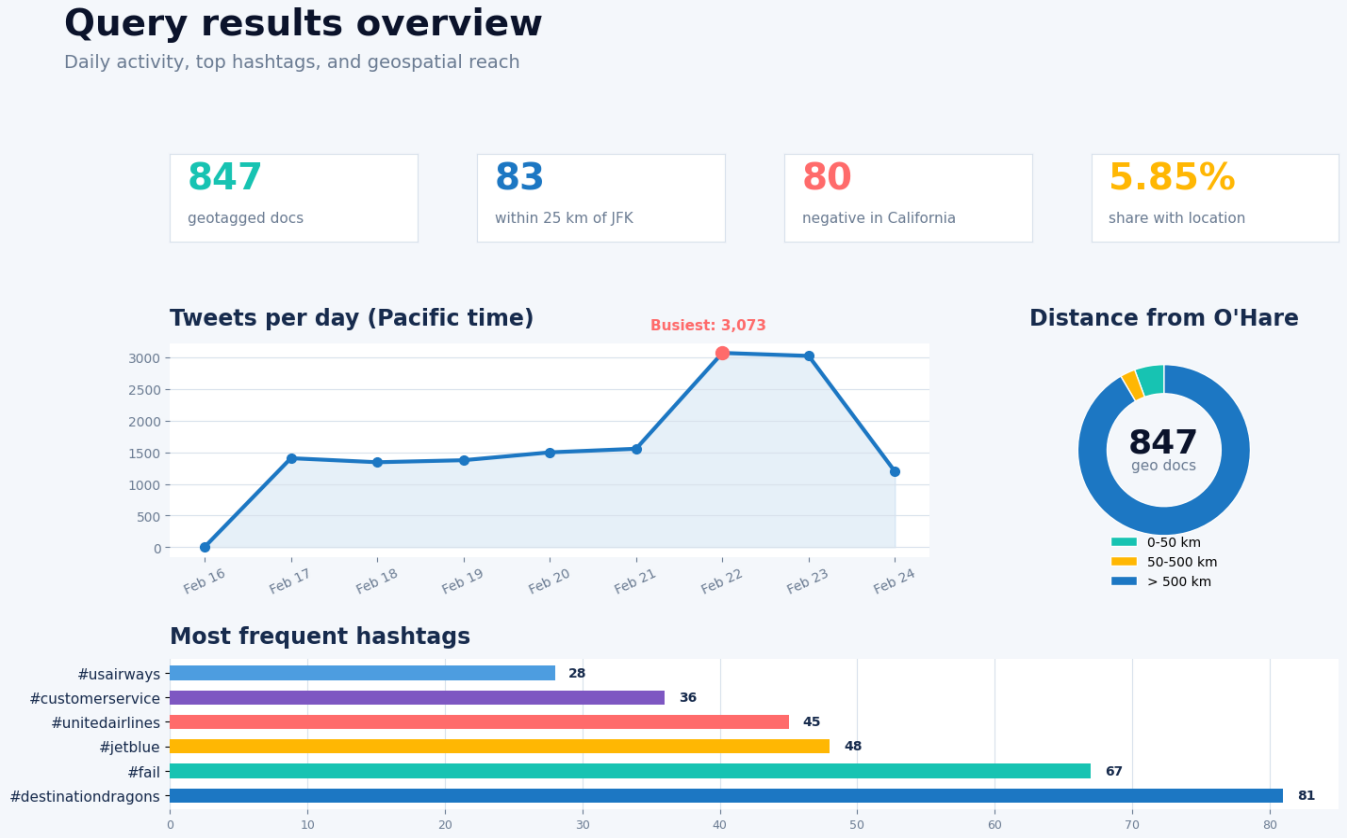


Figure 3. Activity trend, hashtag ranking, and geographic reach.

QUERY	RESULT
Q10	847 of 14,485 documents have a valid location (5.85%).
Q11	83 tweets fall within 25 km of JFK. The nearest is 0.293 km away.
Q12	80 negative geotweets are inside California. United leads with 43.
Q13	O'Hare rings: 47 within 0-50 km, 24 within 50-500 km, and 776 over 500 km.
Q14	Top cell: 6/18/24 with 205 documents. Negative centroid: 39.1647, -89.0527.

The coordinate field is indexed as `geo_point`. Empty, malformed, out-of-range, and [0.0, 0.0] coordinates are omitted so they cannot create false map clusters or distance matches.

Implementation notes

The index is intentionally strict and repeatable. Every design choice supports predictable queries and safe reruns.

Index mapping

FIELD	TYPE	WHY
text	text	English full-text search plus a standard-analyzer subfield.
airline / sentiment	keyword	Exact filters and aggregations.
created_at	date	Ranges and Pacific-time daily buckets.
retweet_count	integer	Numeric ranking.
location	geo_point	Distance, bounding-box, grid, and centroid queries.
hashtags / mentions	keyword	Lowercase exact-value aggregations.

Why Bulk API, tweet_id, and coordinate cleanup?

The Bulk API sends documents in groups of 500, reducing request overhead and making the import faster. Using `tweet_id` as Elasticsearch `_id` makes reruns idempotent: a repeated tweet updates the same document instead of creating a duplicate. The placeholder `[0.0, 0.0]` is removed because it is not a real GPS position.

Method and deliverables

Results were calculated from the complete CSV using the indexer's cleaning rules, including last-write-wins behavior for repeated IDs, Pacific-time day boundaries, the specified geographic limits, and unique lowercase hashtags. The five-row dry-run was executed successfully.

FILE	STATUS
tweets_indexer.py	Completed indexer with strict mapping and bulk loading.
queries.txt	Q1-Q14 in Kibana Console format.
report.pdf	A1, B1, B2, and all query results included.

Dataset: Twitter US Airline Sentiment, CrowdFlower. License: CC BY-NC-SA 4.0.